

Peter Asanov

SENIOR SOFTWARE ENGINEER

Amsterdam, Netherlands

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About

High-impact professional with 20 years' experience, spent over a decade building enterprise-grade distributed systems for real-time interactive and mission-critical applications, navigating complex problems and delivering elegant solutions.

Education

B.Sc. in Computer Science and Engineering

Novosibirsk, Russia

NOVOSIBIRSK STATE TECHNICAL UNIVERSITY (NSTU)

2002 - 2007

Skills

Distributed Systems

architecture & Design, concurrency & memory models, networking, scalability & HA, fault Tolerance & resilience, consensus protocols (Paxos, Raft), CAP Trade-offs, event-driven systems, storage & caching

Dev stacks

Go

gin, zap/slog, gomock

JS/TypeScript

React, Auth.js, tRPC, Prisma, Vite, Tailwind CSS, Axios

Python

FastAPI, uvicorn, SQLAlchemy, celery, paho, pytest/pytest-mock, numpy, pandas, polars, PySpark, MLflow, scikit-learn, seaborn, pyo3/maturin

C++23

boost, flatbuffers, cap'n'proto

Rust

Hyper, Reqwest, clap

JVM

Java, Kotlin, Spring / Boot, Hibernate, NIO

Build Systems

Nix, CMake, ninja, Maven, Meson, Bazel

Cloud Native

IAM/IdPs

AAD(enterprise), RBAC (Azure), IAM, KMS (AWS & GCP), Amazon Cognito, Auth0

Orchestration

Kubernetes (AKS / EKS / GKE), Azure VM, ECS, EC2, GCE

Serverless

Azure Functions, AWS Lambda, GCF, CloudFlare workers

IaC

Terraform, Bicep, Azure Templates, Cloud Formation

CI/CD & GitOps

GitHub Actions, GitLab CI/CD, ArgoCD, Azure DevOps, AWS CodeBuild & CodePipeline

Observability

Azure Monitor / App Insights, AWS CloudWatch, Prometheus, Grafana LGTM, OTLP, Elasticsearch, Jaeger, Zipkin

DevSecOps

Continuous Security, OWASP Top-10, Secure-by-Design, Zero Trust, PoLP, RBAC, Tenant Isolation, Intrusion Detection

Data Systems

DBMS

PostgreSQL, MySQL, Cassandra, MongoDB, QuestDB, CosmosDB, RDS, DynamoDB, GCS, CloudSQL

KV-store

etcd, Redis, RocksDB, Valkey, Memcached

Block storage

S3 / Minio, ABS

Queueing

RabbitMQ, ActiveMQ, Kafka

Big Data

Apache Spark, FLiP (Flink + NiFi + Pulsar), Kafka (Streaming), Airflow, DuckDB

Cryptography

Basics

PKI, Crypto hashes, Symmetric (block) ciphers, KDFs

ECC

ECC / Curve25519, ECDH, EdDSA, Ristretto, Bulletproofs, Pedersen Commitments, Ring Signatures, ZK-SNARKS / STARKS

Work Experience

Senior Software Engineer

Netherlands

FREELANCE / CONTRACTING

2022 -

Contractor / freelancer working on multiple projects, with strong focus on platform engineering and cloud infrastructure:

- Deliverect, full cycle restaurant business automation PaaS: integrated extra dimensions to Ordering facility
Tools: Python, Eve, celery, Kubernetes (GKE), Typescript, React, Redis, minio (local backend)
- Ahold Delhaize (Albert Hijn): online data ingestion platform for internal compliance project, related to R&D efficiency and performance
Tools: Go, Kubernetes (AKS, Kafka Strimzi operator), GitOps (ArgoCD), Terraform
- Private customer: online data ingestion of Taxi terminals data for compliance project to prevent fraud from a driver side
Tools: Go, Kubernetes (AKS), Kafka, Postgres
- LEGO: building internal IAM / IdP functionality
Tools: Azure Functions, Azure SAML identity provider, Python
- Private customer: cloud based video conversion service
Tools: c++20, cmake, ninja, go, grpc-gateway, ffmpeg, nix, terraform(terranix), EKS, CloudWatch
- Private customer: custom video calls service
Tools: c++23, cmake, ninja, asio, libdatachannel (WebRTC stack), nix, ansible
- Petproject [in development]: Pluggable and configuratble HFT matching engine with focus on high performance / ultra-lowlatency. Possible applications: high volume / low latency crypto exchanges, arbitrage bots.
Tools: c++23 with extense usage of SIMD intrinsics, Cap'n Proto, go

Core Software Engineer

Haarlem, Netherlands

AIMMS B.V.

Nov 2019 – Sep 2021

- Centric role in large cloud platform migration project (from AWS to Azure with substantial scratch-rewrite in modern c++, from former Java/Scala)
- Scratch-written Modern c++ framework and related micro-service template featuring protobuf DSL as API first class citizen, popularizing its use also for REST APIs to avoid buggy (back then) OpenAPI generators
- Bootstrapped Azure landing zone (custom bootstrap for Terraform state and RBAC)
- Built scalable tooling and participated in sensitive customer data migration
- Wrote low-code generator of CRUD apps from declarative schema, used for administrative and other internal tools; bringing measurable R&D cost reduction

Tools: ECS, EC2, CloudWatch, Go, Python, Java, Scala, c++17, Typescript, React, conan, cmake, ninja, grpc-gateway, go, Kubernetes (AKS), Azure Application Insights, Terraform

Achievements: tackled cost-efficiency and throughput predictability of compute-heavy optimization workloads (with non-deterministic convergence) by implementing Kubernetes-native solutions: bin packing via resource requests/limits and affinity rules, priority classes, and a pre-warmed spare node strategy using Cluster Autoscaler and taints/tolerations.

Senior Software Engineer

Amsterdam, Netherlands

HVR SOFTWARE (ACQUIRED BY FIVETRAN)

Jan 2018 – Oct 2019

Worked on high performance online data replication system, also called HVR (features and bugfixing)

- Worked on product core (log based replication engine)
- Reversed Oracle histogram data to build custom sharding functionality
- Built a Go REST service for cloud integration.

Tools: C99, Python, Go, multiple RDBMS and Big Data stacks (integration), Oracle (histogram internals), Airflow

Achievements: Rewrote a core feature flag transpiler used in HVR's CI/CD pipeline, cutting runtime from over 1 minute to 5 seconds.

This order-of-magnitude speedup on a critical build path significantly improved developer experience and delivered measurable infrastructure cost savings.

Software Engineer

Eindhoven, The Netherlands

MAPSCAPE B.V.

2017

Worked on Mapscape NDS (navigation data standard) compiler:

- implemented parts of NDS compiler backend related to parsing raw map data
- introduced parallelism in parsing raw map data, significantly speeding it up

Tools: c++ 11 / 14 / boost / gtest / sqlite / shell / python / Airflow

Senior Software Engineer

St. Petersburg, Russia

NEW CLOUD TECHNOLOGIES

2014 - 2017

My Office project: on-prem office solution featuring collaboration engine"

- work on core functionality (features and bugfixing) and on spreadsheets client for Tizen OS

Tools: c++14, cmake, emscripten, Objective C, Qt, SWIG, Kubernetes

Achievements:

- Independently built a cross-platform freetype-based font rendering engine with subpixel positioning for scalable document layouts; overcame technical depth and low project visibility to successfully drive integration into a large enterprise codebase.
- developed advanced spreadsheet UX for Samsung Tizen (embedded Linux), designing a declarative, compile-time state machine to avoid dynamic dispatch overhead and enable precise editing—resolving edge cases, free of bugs present back then in competitors' suites (e.g. Google Docs)

Senior Software Engineer

St. Petersburg, Russia

AIS

2014

Building bank (transaction facility) for crypto mining pool basic exchange functionality:

- scratch-written c++11 / asio framework and a few microservices (and their ownership)
- multi-stage fault tolerant transactions
- operating mission critical distributed system and managing risks of direct loss via using append-only fault-tolerant custom storage, two-factor transactions, impenitent circuit breaking, proper alerting, extensive testing
- zero-to-one release the thing into production as joint effort of 3 engineers exercising high degree of code ownership (on subsystem level)

Tools: c++11, Core Java, Groovy, Ruby(Chef)

Achievements:

- designed and implemented all C++11 microservices for crypto exchange banking facility, including a multi-stage, fault-tolerant transaction system and circuit breaker with (semi-)automatic rollbacks.
- delivered under pressure, a zero-to-one production launch in under 2 months alongside with one more person (Java developer), with personal accountability driving rigorous testing and system reliability under financial risk.

Previous work history

AVAILABLE UPON REQUEST

2005 - 2014